ECA5701 – NANOELECTRONICS

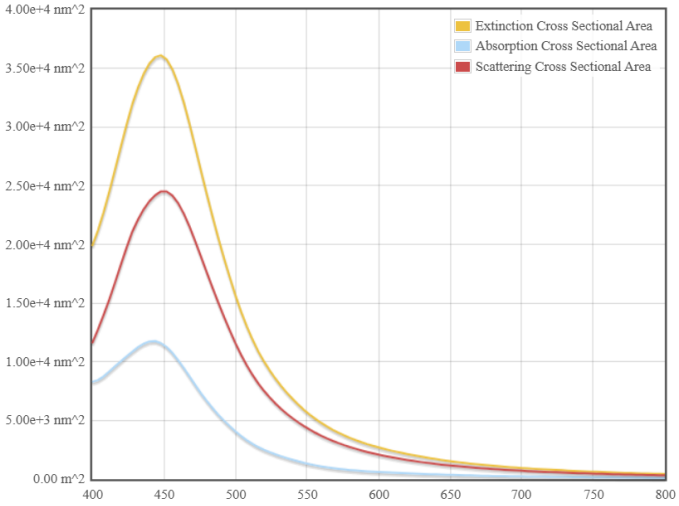
JAYANTH R
192512342

Experiment 1

Simulation of scattering and absorption of multi-layer sphere and coreShell nanoparticle using Mie-plot tool.

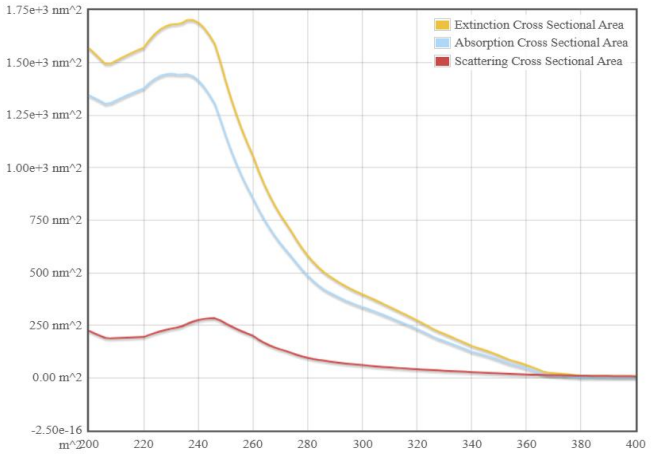
Output:

Core Type	Core RI	Core Diameter (nm)
Gold		20
Shell Type	**Shell RI**	**Shell Thickness (nm)**
Silver		25
Start λ (nm)	**End λ (nm)**	**Medium RI**
400	800	1.33



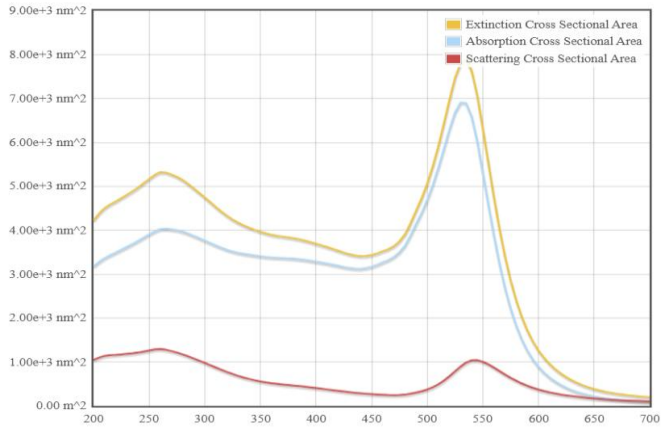
Case 1:

Core Type	Core RI	Core Diameter (nm)
TiO_2		30
Shell Type	**Shell RI**	**Shell Thickness (nm)**
None		0
Start λ (nm)	**End λ (nm)**	**Medium RI**
200	400	1.33



Case 2:

Core Type	Core RI	Core Diameter (nm)
Gold		50
Shell Type	Shell RI	Shell Thickness (nm)
None		0
Start λ (nm)	End λ (nm)	Medium RI
200	700	1.33



Case 3:

Core Type	Core RI	Core Diameter (nm)
Gold		30
Shell Type	Shell RI	Shell Thickness (nm)
Silica		10
Start λ (nm)	End λ (nm)	Medium RI
400	800	1.33

